MIL-F-0051068E(EA)
6 February 1981
SUPERSEDING
MIL-F-0051068D(EA)
4 April 1974
USED IN LIEU OF
MIL-F-51068C
1 December 1970

MILITARY SPECIFICATION

FILTER, PARTICULATE,

HIGH-EFFICIENCY, FIRE RESISTANT

This limited coordination military specification has been prepared by the US Army Armament Research and Development Command based on currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-F-51068C. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers six sizes and seven types of filters for use in air cleaning or air filtering systems (see 6.1).

1.2 Classification. Filters covered by this specification shall be of the following sizes and types (see 6.2).

1.2.1 Sizes.

Size	Nominal rated capacity cubic feet per minute (CFM)
1	25
2	50
3	125
4	500
5	1000
6(M20)	1250

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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1.2.2 Types.

Type	Frame Construction (See 3.2.1.1)
IA	Marine Plywood
IB	Exterior Plywood
IC	Wood Particleboard
IIA	Aluminum Alloy
IIB	Cold Rolled Steel
IIC	Chromized Steel
IID	Stainless Steel

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/11 - Aluminum Alloy 6061, Plate and Sheet.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-698 - Steel, Sheet and Strip, Low Carbon.
QQ-S-766 - Steel Plates, Sheets, and Strip - Corrosion Resisting.
RR-W-370 - Wire Fabric, Steel, Hot-Dipped Galvanized.
TT-E-527 - Enamel, Alkyd, Lusterless.
TT-P-25 - Primer, Paint, Exterior (Undercoat for Wood, Ready-Mixed, White and Tints).
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-C-6183 - Cork and Rubber Composition Sheet; For Aromatic Fuel and Oil Resistant Gaskets.
MIL-S-46055 - Steel, Sheet and Strip, Low Carbon, Chromized.
MIL-F-51079 - Filter Medium, Fire-Resistant, High Efficiency.

STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

DLA26-18-67 - Vibrating Machine, Filter Unit, Q110.
DLB76-2-639 - Penetrometer, Filter Testing, DOP, Q107.
DLB136-14-550 - Tester, Filter, Pressure Resistance, Q160.
DLB136-42-850 - Penetrometer, Filter Testing, DOP, Q76.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TAPPI STANDARDS

T413 - Ash and Paper.

(Request for copies of TAPPI (Technical Association of the Pulp and Paper Industry) Standards should be sent to the Technical Secretary, TAPPI, 1 Donnwoody Park, Atlanta, GA 30341.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standrds

A268 - Seamless and Welded Ferritic Stainless Steel Tubing for General Services.
A366 - Specification for Cold-Rolled Carbon Steel Sheets, Commercial Quality (Tentative).
B209 - Specification for Aluminum Alloy, Sheet and Plate.
D1056 - Sponge and Expanded Cellular Rubber Products (Tentative).
D1225 - Test for Flat Crush of Corrugated Paperboard.
E84 - Surface Burning Characteristics of Building Materials.
A165 - Electrodeposited Coatings of Cadmium on Steel.

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A176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ANSI STANDARDS

ANSI A199.1 - Construction and Industrial Plywood.
ANSI B132.1 - High Efficiency Air Filter Units.

(Copies of these standards may be obtained from American National Standards Institute, 1430 Broadway, New York, NY 10018.)

US DEPARTMENT OF COMMERCE

COMMERCIAL STANDARDS

CS132 - Hardware Cloth.
CS236 - Mat-Formed Wood Particleboard.

(Copies of these standards may be obtained from the US Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Qualification. Particulate filters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of the bids (see 4.2 and 6.4).

3.2 Materials and components.

3.2.1 Frame.

3.2.1.1 Frame material. The frame shall be made from the material as specified for each of the following filter types:

Type IA - 3/4 inch plywood conforming to marine exterior type, grade A-A, of ANSI-A199.1 and fire retardant treated (see 3.2.1.2).

Type IB - 3/4 inch plywood conforming to exterior type, grade A-A, group 1, of ANSI-A199.1 and fire retardant treated (see 3.2.1.2).

Type IC - 3/4 inch fire retardant (see 3.2.1.2) wood particle board for exterior application with filler on both faces and conforming to type 2, grade B (with a minimum density of 45 pounds per cubic foot) class 2 of CS-236.

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Type IIA - Aluminum Alloy 6061-T6, 5052-H32, 5052-H34 or 3003-H14 sheet, 0.064 inch (14 gage B & S) minimum thickness conforming to QQ-A-250/11 or ASTM B209.

Type IIB - Cold-rolled steel sheet, 0.0720 inch (14 gage USS) minimum thickness conforming to QQ-S-698 or ASTM A366. Prior to frame construction, the metal sections shall be cadmium plated in accordance with class 2, type II of QQ-P-416 or ASTM A165, type OS. Welding frame assemblies shall be plated after welding.

Type IIC - Chromized steel sheet, 0.0720 inch (14 gage USS) minimum thickness, conforming to MIL-S-46055.

TYPE IID - Stainless steel, 0.0720 (14 gage USS) minimum thickness, conforming to QQ-S-766 or ASTM A176 with the exception that the chemistry shall be in accordance with TP-409 of ASTM A268.

3.2.1.2 Flame spread classification. The plywood and wood particleboard frames shall have a flame spread classification of 25 or less when tested as specified in ASTM E84.

3.2.2 Gasket material. Unless otherwise specified in the contract or purchase order, (see 6.2), gasket material shall be one of the following materials:

(a) Cork and rubber composition conforming to type I or type II, class 2, grade A, soft of MIL-C-6183.

(b) Oil resistant, expanded cellular rubber in accordance with grade SCE-43 or 44 of ASTM 1056 and also qualified for listing on the Qualified Products List (QPL) for type II, grade A, medium of MIL-R-6130.

3.2.3 Faceguards. Unless otherwise specified in the contract or purchase order, faceguards shall be fabricated from galvanized 4 by 4 mesh, wire fabric (hardware cloth) made from 23 gage steel wire conforming to RR-W-370, or hardware cloth conforming to CS 132.

3.2.4 Adhesives and sealants. Adhesives used to splice the medium, fasten gasket to filter face, and sealants or adhesives used to seal the filter pack to the frame shall be self-extinguishing after direct contact with an open flame, as evidenced by testing in accordance with the spot-flame test of ANSI B132.1 (see 6.3).

3.2.5 Filter medium. Unless otherwise specified in the contract or purchase order, the filter medium shall conform to MIL-F-51079.

3.2.6 Separators. Unless otherwise specified in the contract or purchase order the pleats of the filter medium shall be spaced by corrugated separators made of the following materials (see 6.2):

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Aluminum - 0.0015 inch thick, minimum, conforming to ASTM B209, Alloy, 5052-H39, 3003-H19, or 1145-H19, having a minimum resistance to compression of 10 pounds when tested in accordance with ASTM D 1225 (see 6.7).

Asbestos - having a minimum density of 4 pounds per 100 square feet, water repellant, shall not support combustion when ignited in still air, having a maximum of 22 percent combustibles when tested in accordance with TAPPI Standard T413, modified to 1000°C for 5 minutes; and when tested in accordance with ASTM D1225 (see 6.7) shall have a minimum resistance to compression of 25 pounds dry, and 8 pounds wet, after soaking under 5 inches of water at 25 ± 5°C for 1 hour.

3.2.7 Paint finish for filter frame. Where required (see 3.3.7) the exterior surface of the frame shall receive one coat or primer conforming to TT-P-25, tinted to grey matching color No. 26134, Fed. Std. 595 and two coats of olive drab, No. X-34087 finish conforming to TT-E-527.

3.3 Manufacture and assembly.

3.3.1 Construction. Unless otherwise specified in the contract or purchase order each size of filter shall be constructed to conform to figure 1 (see 6.2). The dimensions for the various filters shall be as specified in table I. The fasteners used in the construction of the filter frames shall be compatible with frame materials (see 3.4.7).

3.3.2 Flatness and squareness. The faces of the frame shall be flat and parallel to within a total allowance of 1/16 inch. The frame shall be square to within a total allowance of 1/8 inch when measured diagonally acrosss the corners of both faces.

3.3.3 Gasket assembly. Unless otherwise specified in the contract or purchase order the gasket shall be fastened to the influent face of the filter frame, and shall be made from the material specified in the contract or purchase order (see 6.2). The gasket shall be 1/4 ± 1/32 inch thick by 3/4 ± 1/32 inch wide and flush with the outer edges of the frame. The gasket shall be either molded in continuous unbroken form, or made from 4 strips joined at the corners by interlocking means, so that no gaps are visible and the joint shall be air tight. The gasket shall be continuously cemented to the face of the filter frame so as to prevent any air leakage between gasket and frame.

3.3.4 Faceguards. Unless otherwise specified in the contract or purchase order faceguards shall be used only on filter sizes 4, 5, and 6 (see 6.2). A faceguard shall be installed in each face of the filter in such a manner as to be essentially flat.

3.3.5 Medium installation. The filter medium shall be securely fastened to sides and ends of filter frame with adhesive to completely seal edges of medium to filter frame. Patching of holes or tears in the medium shall not be permitted.

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3.3.5.1 Splices. Splicing of the medium shall not be permitted in sizes 1 and 2. Larger size filters shall not have more than one splice in the filter pack.

TABLE I. Filter dimensions

Filter size	Dimensions, inches		
	X	Y	Z*
1	8 (+0 - 1/16)	8 (+0 -1/16)	3-1/16 (+1/16 -0)
2	8 (+0 - 1/16)	8 (+0 -1/16)	5-7/8 (+1/16 -0)
3	12 (+0 - 1/16)	12 (+0 -1/16)	5-7/8 (+1/16 -0)
4	24 (+0 - 1/8)	24 (+0 -1/8)	5-7/8 (+1/16 -0)
5	24 (+0 - 1/8)	24 (+0 -1/8)	11-1/2 (+1/16 -0)
6	24 (+0 - 1/8)	24 (+0 -1/8)	11-1/2 (+1/16 -0)

*Excluding the gasket, no part of the filter shall extend beyond this dimension.

3.3.6 Marking. Each filter shall be marked by stencilling or with a non-removable label on the top of the frame to indicate the manufacturer, model, serial number, test airflow direction, airflow resistance at rated airflow and the results of the dioctylphthalate (DOP) smoke penetration test at rated airflow and at 20 percent of rated airflow for all sizes of filters. In addition the wording "Install With This Side Up" shall be stencilled on the top of the frame in 3/8 inch minimum height letters. The top of the filter shall be perpendicular to the pleats in the filter medium (see figure 1).

3.3.7 Finish. The exterior surface of type I (wood) size 6 filter frames shall be painted as specified in 3.2.7. Unless otherwise specified in the contract or purchase order, sizes 1, 2, 3, 4 and 5 shall be unpainted.

3.4 Performance.

3.4.1 DOP smoke penetration. The total DOP smoke penetration through the filter medium, frame, and gasket which has been encapsulated, shall be no greater than 0.03 percent of upstream concentration at rated aiflow and at 20 percent of rated airflow for all sizes when tested as specified in 4.3.4.1.

3.4.2 Resistance to airflow. The resistance to airflow at the rated airflow of the filter shall be no greater than 1.0 inch of water for filters, sizes 1, 2, 3, 4 and 5, and 1.25 inches of water for size 6 filters when tested as specified in 4.3.4.1.

3.4.3 Resistance to rough handling. The filter shall be free from cracked or warped frames; loose corners or joints; cracked adhesive; and loose or

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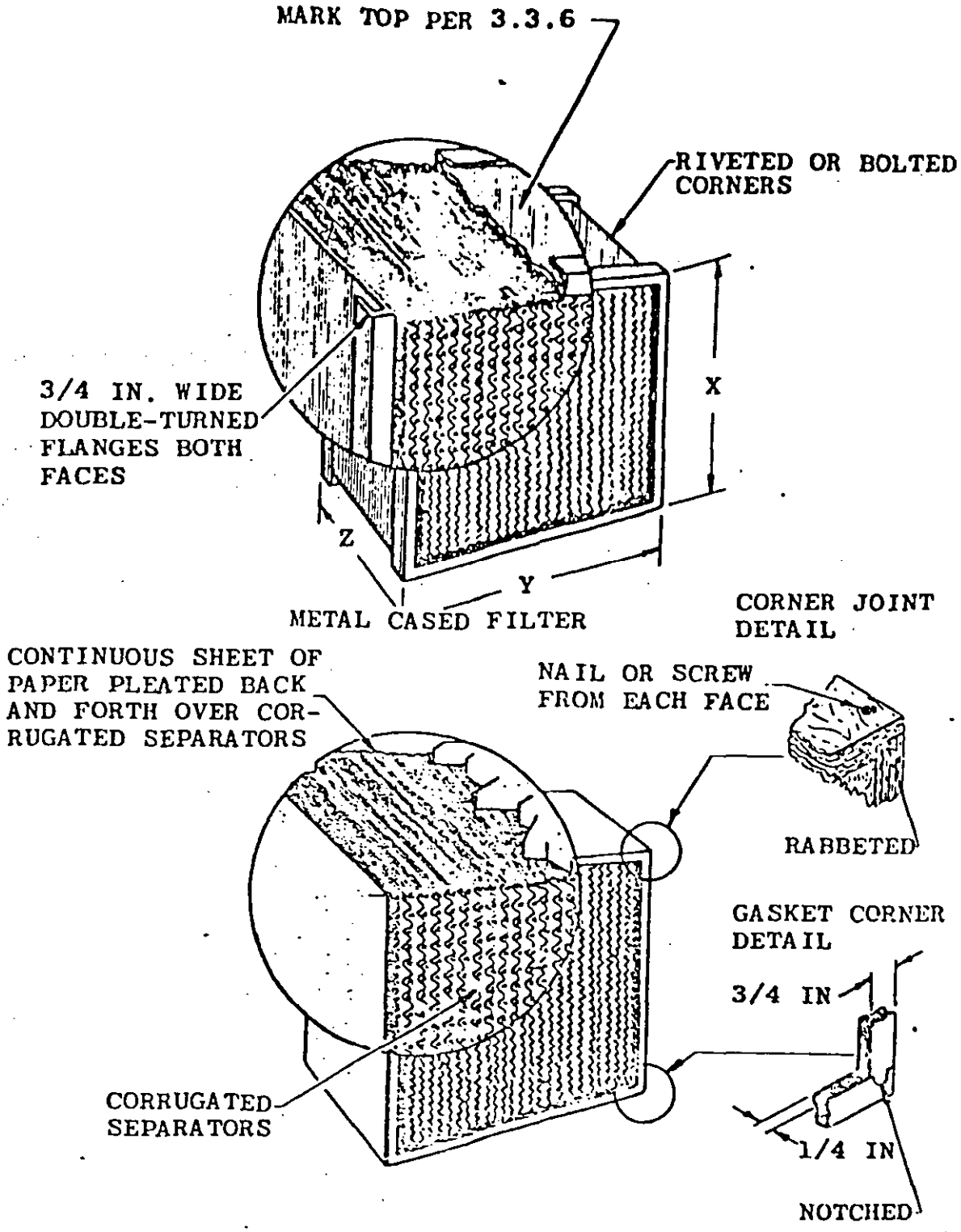


FIGURE 1

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deformed medium, separators, or faceguards after rough handling as specified in 4.3.4.2 for 15 minutes at 3/4 inch total amplitude and a frequency of 200 cycles per minute with faces, pleats, and separators in a vertical position (see figure 1). Also after rough handling the same filter shall meet the requirements in 3.4.1 and 3.4.2.

3.4.4 Resistance to pressure. Filters sizes 4, 5 and 6 after being conditioned in accordance with 3.4.4.1 shall withstand the airflow and water spray environment of 3.4.4.2 without rupture of the filter medium; and within 15 minutes after completion of the pressure test and still wet the filter shall meet the DOP requirement of 3.4.1 at 20 percent airflow.

3.4.4.1 Conditioning. Prior to being tested for resistance to pressure, the filter shall be conditioned for 24 hours minimum in a chamber at 95° ± 5°F and a relative humidity (RH) of 95 ± 5 percent.

3.4.4.2 Pressure testing. After conditioning as specified in 3.4.4.1, the filter shall be removed from the chamber, wrapped, transported to the test equipment, and tested within 15 minutes in accordance with 4.2.3.1 under the test conditions listed in table II.

TABLE II. Test conditions and requirements

Test conditions	Test requirements
Temperature	95° + 5°F
Relative humidity	95 + 5%
Rate of airborne water droplets flowing toward the filter	1 ± 1/4 pound per minute per 1000 cfm of nominal rated filter capacity
Pressure differential across filter	10.0 ± 0.2 inch of water
Time to reach pressure differential	1/2 minute, maximum
Time duration at sustained water droplet flow and pressure differential across filter	1 hour, minimum
Air flow	That required for producing the above pressure differential

*This is defined as the rate of water flowing through the spray orifice less the fallout and drainage from the air duct walls between points of location of the spray orifice and one inch before the face of the filter.

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3.4.5 Resistance to heated air. The DOP smoke penetration through the filter medium, frame and gasket, when tested at the rated airflow of the filter and in accordance with 4.3.4.1 shall not exceed 3.0 percent after heated air at a temperature of 700° ± 50°F has been passed through the filter (face of gasket and sides of filter not exposed) for no less than 5 minutes at the airflow rate for size specified in table III and tested in accordance with 4.2.3.2.

TABLE III. Airflow rate for heated air

Filter Size	Minimum Test Airflow at 700° ± 50°F (cfm)
1	25
2	50
3	125
4	400
5	1000
6	1250

For sizes 1, 2, 3, 4, and 5 (table III) and Underwriter's Laboratories (UL) label shall be acceptable objective evidence for compliance with this requirement when tested in accordance with the applicable test method in ANSI B132.1.

3.4.6 Spot flame resistance. After the removal of the test flame at each point of application, there shall be no sustained flaming on the downstream face of the unit and there shall be no transmittal of flame to the outside surfaces of the frame, gaskets, separators, medium or adhesive when tested in accordance with 4.2.3.3.

With filter sizes of 1, 2, 3, 4 or 5, a UL label shall be acceptable objective evidence of compliance with this requirement when tested in accordance with the applicable test method in ANSI B132.1.

3.4.7 Resistance to environmental exposure. The filters shall show no evidence of corrosion of metal frames, fasteners or hardware cloth after any cycle and shall comply with the requirements specified in 3.4.7.3 when subjected to the environmental conditions, cyclic exposures, and tested in accordance with 3.4.7.1, 3.4.7.2 and 3.4.7.3.

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3.4.7.1 Environmental conditions. The filters shall be subjected, in accordance with 4.2.3.4 to three 3-week cycles of environmental conditions each consisting of one week arctic (-65°F) one week desert (160°F and 10 percent RH) and one week tropical (113°F and 88 percent RH). The time between condition changes shall not exceed 15 minutes. The sequence of environmental changes for each cycle and the number and sequence of cycles shall be as specified in 3.4.7.2.

3.4.7.2 Cyclic exposure. The filters shall be subjected to 3 cycles (a total duration of 9 consecutive weeks) varied as follows:

Cycle 1 - Tropical, arctic desert
Cycle 2 - Arctic, desert tropical
Cycle 3 - Desert, tropical, arctic

After each cycle the filters shall meet the requirements specified in 3.4.7.3.

3.4.7.3 Testing after exposure cycles. The filters shall meet the DOP penetration requirements of 3.4.1 and the airflow resistance of 3.4.2 after cycle 1 and again after cycle 2. After cycle 3 the filters shall meet the rough handling requirements of 3.4.3.

3.5 Workmanship. The filter shall be free from foreign matter (dirt, oil or viscous material) and damage, such as distorted or cracked frame, deformation or sagging of medium, separators and faceguards, cracks in adhesive and cracks and holes in exposed portions of the medium.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Government's responsibility. Unless otherwise specified, the Government will be responsible for the performance of the tests in 4.2. Samples will be forwarded to the laboratory designated by the contracting officer (6.2).

4.1.3 Objective evidence. The contractor shall provide objective evidence including UL labels acceptable to the contracting officer that the requirements of 3.2, 3.4.5, 3.4.6 and section 5 for which specific inspection has not been provided in the specification have been satisfied.

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4.2 Qualification.

4.2.1 Sample. Except as noted below qualification sample of 11 each of size 5 or 6 particulate filters as specified shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production (see 6.2). Where applicable, UL labels shall be affixed to the filters as evidence of conformance with 3.2.1.2, 3.4.5 and 3.4.6 prior to submission of the filters for qualification. Qualification of size 6 (M20) filter shall be extended to sizes 1 through 5 filters of the same type, materials and source of manufacturer. Qualification of size 5 filter shall be extended to sizes 1 through 4 filters of the same type, materials and source of manufacturer.
NOTE - A qualification sample is not required for those filters which were previously qualified and appear in the applicable Qualified Products List, or were tested and approved for inclusion in the list prior to the bid opening date.

4.2.1.1 Samples with splice. If the contractor intends to deliver filters with a splice in the filter medium (3.3.5.1) three of the qualification samples shall contain a splice.

4.2.1.2 Filters with splice. Filters containing a splice shall be clearly marked with the words 'spliced medium' printed with 1/2 inch minimum height letters and located 2 inches below the marking cited in 3.3.6.

4.2.1.3 Material change. Except as noted below, any change in materials or source of manufacturer shall require a new qualification sample (4.2.1).

NOTE - When a qualified manufacturer changes his source of filter medium, a new qualification sample shall be required except when the new medium is of the same grade as that used for qualification, and appears in the applicable medium Qualified Products List, or has been tested and approved for inclusion in the list prior to the bid opening date.

4.2.1.4 Reverification of qualification. At least once every 5 years, each manufacturer on the Qualified Products List shall submit samples in accordance with 4.2.1 for requalification.

4.2.2 Inspection.

4.2.2.1 For examination. Each filter in the qualification sample shall be examined for all the characteristics in the classification of defects (4.3.3.3).

4.2.2.2 For tests. The qualification sample shall be tested for all the requirements of this specification in accordance with the groupings, sizes of filter, quantity and sequence of tests shown in table IV. If applicable (4.2.1.1) one filter in each group A, B and C shall contain a splice and so indicated.

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TABLE IV. Test groups and sequence

Group Quantity Size	Test sequence Requirement	Test paragraph
Group - A Quantity - 4 filters Sizes - 5 or 6	Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow and at 20% of rated airflow (3.4.1) Resistance to pressure (3.4.4) DOP smoke penetration at 20% of rated airflow only (3.4.1)	4.3.4.1 4.3.4.1 4.2.3.1 4.3.4.1
Group - B Quantity - 4 filters Sizes - 5 or 6	Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow and at 20% of rated airflow (3.4.1) Resistance to rough handling (3.4.3) Resistance to heated air (3.4.5) DOP smoke penetration at rated airflow only (DOP penetration requirement from 3.4.5)	4.3.4.1 4.3.4.1 4.3.4.2 4.2.3.2 4.3.4.1
Group - C Quantity - 2 filters Sizes - 5 or 6	Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow and at 20% of rated airflow (3.4.1) Resistance to environmental exposure (3.4.7) Resistance to rated airflow at end of cycle 1 and cycle 2 (3.4.2) DOP smoke penetration at rated airflow at end of cycle 1 and cycle 2 (3.4.1) After third cycle: Resistance to rough handling (3.4.3) Examination for damage (3.4.3) Resistance to rated airflow (3.4.2) DOP smoke penetration at rated airflow and at 20% rated airflow (3.4.1)	4.3.4.1 4.3.4.1 4.2.3.4 4.3.4.1 4.3.4.1 4.3.4.2 4.3.4.2 4.3.4.1 4.3.4.1
Group - D Quantity - 1 filter Sizes - 5 and 6	Resistance to spot flame (3.4.6)	4.2.3.3

4.2.3 Tests.

4.2.3.1 Resistance to pressure. The resistance to pressure for sizes 4, 5 and 6 filters shall be determined with the Q160 Pressure Resistance Tester (DLB 136-14-550).

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4.2.3.2 Resistance to heated air. The resistance to heated air for all sizes of filters shall be determined using a single pass air preheater capable of delivering air at test airflow (table III) and temperature as specified in 3.4.5. In addition, the temperature differential across the face of the filter shall not vary by more than ± 15°F from the mean. After the hot air chamber has reached 700° ± 50°F, the filter shall be placed into the chamber, and when the chamber temperature recovers to 700°F ± 50°F, the 5-minute exposure shall commence (see 6.5).

4.2.3.3 Resistance to spot flame. The resistance to spot flame shall be determined in accordance with the applicable test method with ANSI BI32.1.

4.2.3.4 Resistance to environmental exposure. The filter shll be exposed in an environmental chamber to the conditions specified in 3.4.7. After cycle 1 and again after cycle 2, the DOP smoke penetration and resistance to airflow shall be determined in accordance with 4.3.4.1. At the end of the third cycle, the filter shall be rough handled and examined for damage in accordance with 4.3.4.2 and the DOP penetration determined in accordance with 4.3.4.1.

4.2.4 Acceptance/rejection criteria.

4.2.4.1 Qualification examination and tests. Failure of any filter to comply with the requirements of this specification when examined and tested as specified in 4.2.2 shall disqualify the entire sample.

4.3 Quality conformance inspection.

4.3.1 Lotting. A lot shall consist of the completed filters of the same size produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of filter medium shall be represented in any one lot of finished filters.

4.3.2 Sampling for examination and tests. Sampling for examination and tests shall be conducted in accordance with MIL-STD-105. For major defect 101, resistance to rough handling, level S-4 shall be used.

4.3.3 Inspection procedure.

4.3.3.1 For examination and test. Sample filters shall be examined and tested in accordance with the classification of defects (4.3.3.3) and with MIL-STD-105.

4.3.3.2 For critical defects. Each filter in the lot shall be tested for its critical characteristics listed in the classification of defects.

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4.3.3.3 Classification of defects.

(a) Filter, particulate, high-efficiency, fire resistant (figure 1).

Categories	Defects	Acceptance standards
Critical:		
1	DOP penetration (3.4.1)	4.3.4.1
2	Resistance to airflow (3.4.2)	4.3.4.1
Major:	AQL 1.5 percent defective	
101	Resistance to rough handling (3.4.3)	4.3.4.2
	AQL 1.0 percent defective	
102	Component incorrect, missing, or incorrectly assembled (3.3.1)	
103	Dimensions not as specified (3.3.1)	
104	Damaged to exposed portions of medium (3.4.3)	
105	Flatness incorrect (3.3.2)	
106	Faceguard incorrect (3.2.3), (3.3.4)	
107	Frame not square within 1/8 inch on diagonal (across face) (3.3.2)	
108	Frame faces not parallel within 1/16 inch (3.3.2)	
109	Filter pack shows patches or more than one splice (3.3.5.1)	
110	Finish not as specified (3.3.7)	
111	Gaskets not as specified (3.3.3)	
112	Workmanship (3.5)	
Minor:	AQL 2.5 percent defective	
201	Faceguard not essentially flat (3.3.4)	
202	Marking missing or illegible (3.3.6)	
203	UL label missing (3.4.5, 3.4.6)	

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(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Marking missing, incorrect or illegible
102	Filter not oriented properly in box
103	Filter face not covered as specified

4.3.4 Tests.

4.3.4.1 Resistance to airflow and DOP smoke penetration. The resistance to airflow and DOP smoke penetration for sizes 1, 2 and 3 filters shall be determined with the Q76 DOP Penetrometer (DLB 136-14-850). The resistance to airflow and DOP smoke penetration for sizes 4, 5 and 6 filters shall be determined with the Q107 DOP Penetrometer (DLB 76-2-639).

4.3.4.2 Resistance to rough handling. The filters shall be rough handled with the Q110 Vibrating Machine, DLA 26-18-67), examined for damage and the DOP penetration determined in accordance with 4.3.4.1. Filters shall be tested without packing or packaging.

4.3.5 Removal from the Qualified Products List. If at any time the Government elects, as set forth in 4.1.1, to test any filter procured under this specification against any of the requirements of this specification, and the filter fails to meet the requirements as determined by Chemical Systems Laboratory (DRDAR-CLW-E) the manufacturer will be notified and removed from the applicable Qualified Products List.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified herein (see 6.2).

5.1.1 Level A (size 6 filters M20). The M20 filters shall be preserved and packaged as specified on the repair parts packaging data sheet which is identified by the National Stock Number.

5.1.2 Commercial (size 1 through 6 filters). Filters of one size (see table I) shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A or commerical (see 6.2).

5.2.1 Level A (size 6 filter [M20]). Each filter packaged as specified in 5.1.1 shall be packed with the filter pleats and separators in the vertical position within a fiberboard box conforming to style RSC, grade V3c (for a type 1 load) of PPP-B-636. Closure shall be in accordance with requirements for

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closure of class weather-resistant boxes in the appendix of the box specification. In addition, boxes shall be taped in accordance with the requirements for taping style RSC, class weather-resistant boxes to be shipped as exterior containers; and shall be reinforced with tape, in accordance with the applicable requirements in the appendix of the box specification.

5.2.2 Commercial (size 1 through 6 filters). Filters of one size shall be packaged as specified in 5.1 and packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special markings required by the contract or order, marking shall be as specified herein.

5.3.1 Level A packaging. (5.1.1) shall be marked in accordance with the repair parts packaging data sheet which is identified by the National Stock Number. Level A shipping containers (5.2.1) shall be marked in accordance with MIL-STD-129 and shall include the special marking of 5.3.3.

5.3.2 Commerical packaging. (5.1.2) and packing (5.2.1) shall be marked in accordance with MIL-STD-1188 and shall include the special marking of 5.3.3.

5.3.3 Special marking. The commercial unit containers (5.1.2) and shipping containers (5.2.2) shall be clearly marked with the following information:

(a) Mark on two adjacent sides the word "UP" with an arrow toward the top of the container.

(b) Mark the vertical position of the pleats.

(c) "FRAGILE" and "HANDLE WITH CARE" labels and/or markings shall be applied to two adjacent sides of the box.

5.4 Repair part (size 6 filter M20). When this item is procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. Each filter covered by this specification is intended to be used as a particulate filter, singly or in parallel in air purifiers or air filtering equipment that will be used in air at temperatures up to 250°F and which may contain entrained moisture.

6.2 Ordering data. Procurement documents should specify:

(a) The title, number and date of this specification.
(b) Filter size, 1.2.1.
(c) Filter type, 1.2.2.
(d) Gasket material, 3.2.2; and location if other than specified in 3.3.3.

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(e) Faceguard requirement if other than specified in 3.3.4, and material 3.2.3.
(f) Construction if other than specified in 3.3.1.
(g) Finish if other than specified in 3.3.7.
(h) The level of packaging and packing.
(i) The laboratory to which samples are to be sent for testing.
(j) Filter medium, if other than specified in 3.2.5.
(k) Separator requirement, if other than specified in 3.2.6.
(l) Quantity of packages per shipping container.
(m) Separator material (3.2.6).

6.3 Adhesives and sealants. Any adhesive or sealant found suitable during the qualification tests will be acceptable.

6.4 Qualification. With respect to products requiring qualifications, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders on the products covered by this specification. Information pertaining to qualification may be obtained from Commander, US Army Armament Research and Development Command, ATTN: DRDAR-CLW-E, Aberdeen Proving Ground, MD 21010. Samples shall not be shipped until the above addressee has furnished written shipping instruction to the manufacturer.

6.5 Heated air tester. A suitable tester which is currently used at Edgewood Arsenal consists of the following components connected in series in the order given: an air supply fan (1500 CFM @ 700°F, 4" S.P.), a gas fired air heater (700,000 Btu/hr), an air mixer (baffles and louvers), a test chamber, and an exhaust duct. Liquid propane gas (ca. 2500 Btu/ft.^{3}) is used as a fuel for the heater and the exhaust is vented to the outside. Both the airflow and the temperature controls are adjustable. The holding fixture for the test filter shall permit all the air to pass through the filter and shall prevent the air from passing over the face of the filter gasket and sides of the filter. This tester is available for inspection at Product Assurance Directorate, US Army Armament Research and Development Command, Edgewood Area, Aberdeen Proving Ground, MD 21010.

6.6 Interplant shipment. Packaging for supplies and materials that do not enter the military supply system. Typical are shipments from a vendor to a contractor, a sub-contractor, or a contractor to a military arsenal or plant.

6.7 Compression testing. Suitable test equipment for measuring resistance to compression is: Instron Model 1101 tensile tester equipped with a variable speed range, Instron number 2511-202 compression load cell, 50 pounds maximum, Instron number 2501-025 compression anvil and cable, Instron number A6-17,

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6-inch diameter compression plate, coupled to a high speed graphic recorder. Also, a 4-inch square sample has been found to be suitable for performing this test.

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Army - EA

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